GRAINGUARD · USER REQUIREMENTS

GrainGuard user requirements

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-24	Amish Chadha	First issue (kit 1.8 upgrade), drawn from the users, constraints and decisions in GGD-PRB-001 and GGD-DDR-001

GrainGuard user requirements

What the people who store grain, fit the kit and live with it need, written from their side. Seventeen needs, twelve of them High. The product requirements that meet them are in GGD-PRS-001 (`03-requirements.md`) and the links are in the traceability matrix GGD-RTM-001 (`03b-traceability.md`). The needs come from the users and stakeholders in GGD-PRB-001 (Tables 1 and 2) and from Amish's decisions in GGD-DDR-001; none has yet been checked with a farmer, so all are proposals for the first co-design visit (GGD-DEC-001, item 4 of 2026-10-02). Method: STANDARDS section 22.

Table 1. User requirements.

ID	Category	Title	Need	Priority	Source
UR-001	Performance	Grain temperature through the depth	I want to know the temperature of my grain from the floor to the top without climbing the bin or entering it	High	Small or mid-size grain farmer (PRB Table 1)
UR-002	Performance	Drying or rewetting	I want to know whether my grain is drying or taking on water while it is stored	High	Small or mid-size grain farmer
UR-003	Performance	Outside air at the bin	I want the controller to judge the air the fan actually pulls in, at my bin, not a forecast	Medium	Small or mid-size grain farmer; extension specialist (PRB Table 2)
UR-004	Performance	Fans run only when they help	I want the fan to run when the air will cool or dry the grain, and to stay off when it would rewet or rewarm it	High	Small or mid-size grain farmer; extension specialist
UR-005	Safety	Safe when the controller fails	If the controller dies or loses its readings, the fan must stop, and I must still be able to run it by hand	High	Farm electrician; small or mid-size grain farmer
UR-006	Performance	Warning of heating	I want to be warned early when part of the grain starts to heat, before it spoils	High	Small or mid-size grain farmer
UR-007	Usability	Status from the farmhouse	I want to see how the bin is doing and be alarmed in the house, without walking out to the bin	Medium	Farm worker or family member
UR-008	Environment and durability	No mains, all winter	It must keep working at a bin with no outlet nearby, through short winter days and cloudy weeks	High	Small or mid-size grain farmer
UR-009	Environment and durability	Survives the bin's conditions	It must survive a northern winter, a hot dusty summer and a fumigation of the bin	High	Small or mid-size grain farmer; licensed fumigator (PRB Table 2)
UR-010	Safety	No damage to the roof	Whatever hangs in the bin must not overload the roof or break when the grain is unloaded	High	Bin maker (PRB Table 2); small or mid-size grain farmer
UR-011	Safety	No entry into a bin holding grain	Nobody should ever have to enter a bin holding grain to fit, check or service it	High	Farm worker or family member
UR-012	Safety	Mains work kept small	The only mains work should be a small, clearly isolated job an electrician can do at the starter	High	Farm electrician
UR-013	Cost	Affordable per bin	It must cost far less per bin than the commercial monitoring and fan control systems	High	Small or mid-size grain farmer; cooperative or storage manager
UR-014	Regulatory			Medium	

ID	Category	Title	Need	Priority	Source
		Data stays on the farm	My readings stay on my farm and need no cloud account		Small or mid-size grain farmer
UR-015	Deployment and assembly	Retrofit, workshop built	It must fit my existing bin and fan with no welding or cutting of bin sheets, and be buildable in a farm workshop	Medium	Small or mid-size grain farmer; open hardware community (PRB Table 1)
UR-016	Maintenance	Repairable	I want to repair it with bought parts and the documents, without the maker	Low	Open hardware community; cooperative or storage manager
UR-017	Regulatory	Legal radio	The radio must be legal to use where the farm is	Medium	Radio regulator (PRB Table 2)

Categories: deployment and assembly, performance, usability, safety, environment and durability, logistics, maintenance, cost, regulatory. Priorities: High, Medium, Low.